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Intro

Phd Thesis Review

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Interactive Mapping Application for Environmental Monitoring and Forecasting

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1. Introduction

Air pollution is a major environmental threat to ecological sustainability and human health across the globe. Pollutants such as Nitrogen Dioxide (NO_2), Ozone (O_3), Particulate Matter (PM10) and Sulphur Dioxide (SO_2) are known to cause respiratory illnesses, heart and blood vessel disease, and damage to the environment. Therefore, monitoring and analysis of air quality data is crucial to understand air pollution patterns and help guide good environmental management practices. DEFRA maintains a comprehensive air quality monitoring network in the UK. These monitoring stations are gathering pollutant data continuously in various regions, generating huge amounts of environmental data. This data can be used to gain valuable insights into the trend of pollution, seasonal changes, and geographical differences in air quality.

This project is to create an air quality prediction and hotspot detection system based on the concentration of Nitrogen Dioxide (NO_2). The process of predictive modelling requires a thorough exploration of the data structure and characteristics, which is achieved by the process of exploratory data analysis (EDA). Hence, this Technical Progress Report is an account of the process of dataset collection, an overview of the dataset and an exploratory analysis of the air quality data. The results achieved in this step will be used to guide the following pre-processing, feature engineering, machine learning model building and hotspot detection steps.

2. Dataset Collection

2.1 Primary Air Quality Dataset

The main data used in this study were sourced from the DEFRA UK-AIR (Department for Environment, Food and Rural Affairs – United Kingdom Air Information Resource) platform. The data includes air quality data from monitoring sites spread throughout the United Kingdom. Data contains information about several pollutants, monitoring sites, time and site properties. Ten main attributes can be found in the dataset: StartTime, EndTime, Status, Unit, Value, PollutantName, SiteName, SiteType, Region and Country. A record is a data sample taken from a monitoring station at a particular monitoring time and for a given pollutant. The pollutants are: Nitrogen Dioxide (NO_2), Ozone (O_3), Particulate Matter (PM10), Sulphur Dioxide (SO_2). Nitrogen Dioxide (NO_2) has been chosen as the main pollutant of interest because it is a major air quality pollutant and a public health issue. NO_2 levels are typically high in traffic, industrial and urban pollution areas. So, the key objective of this research is the prediction of NO_2 contents is the main objective.

```python
import pandas as pd

df_main = pd.read_csv('airQuality_2024 data.csv')
df_main.head()
```

	StartTime	EndTime	Status	Unit	Value	PollutantName	SiteName	SiteType	Region	Country
0	2024-12-31T23:00:00Z	2024-12-31T24:00:00Z	V	ugm-3	7.63778	Nitrogen dioxide	Derby St Alkmund's Way	UrbanTraffic	East Midlands	England
1	2024-12-31T22:00:00Z	2024-12-31T23:00:00Z	V	ugm-3	7.22299	Nitrogen dioxide	Derby St Alkmund's Way	UrbanTraffic	East Midlands	England
2	2024-12-31T21:00:00Z	2024-12-31T22:00:00Z	V	ugm-3	10.05369	Nitrogen dioxide	Derby St Alkmund's Way	UrbanTraffic	East Midlands	England
3	2024-12-31T20:00:00Z	2024-12-31T21:00:00Z	V	ugm-3	7.8517	Nitrogen dioxide	Derby St Alkmund's Way	UrbanTraffic	East Midlands	England
4	2024-12-31T19:00:00Z	2024-12-31T20:00:00Z	V	ugm-3	11.34749	Nitrogen dioxide	Derby St Alkmund's Way	UrbanTraffic	East Midlands	England

```python
df_main.head()
df_main.shape
df_main.columns
df_main.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1048575 entries, 0 to 1048574
Data columns (total 10 columns):
 #   Column         Non-Null Count    Dtype
---  ------         --------------    -----
 0   StartTime      1048575 non-null  object
 1   EndTime        1048575 non-null  object
 2   Status         1048575 non-null  object
 3   Unit           1048575 non-null  object
 4   Value          1048575 non-null  object
 5   PollutantName  1048575 non-null  object
 6   SiteName       1048575 non-null  object
 7   SiteType       1048575 non-null  object
 8   Region         1048575 non-null  object
 9   Country        1048575 non-null  object
dtypes: object(10)
memory usage: 80.0+ MB
```

Figure 1: Initial Dataset Preview and Structure Analysis of Air Quality Data Using Pandas

Observation

The data has 10 attributes and 1,048,575 records. The first look at the data revealed that pollutant data are available from several monitoring stations in various parts of England.

2.2 Secondary Geographic Dataset

Kaggle was used to collect a secondary dataset for use in the spatial analysis part of the project. The data used were the UK DEFRA AURN air quality dataset (2015-2023), which included air quality and geographic data for monitoring stations. Key characteristics of this dataset are monitoring station name, latitude, longitude, site type, and pollutant data. The data set includes several environmental variables, but it was necessary only for geographic attributes to be used in this study. The main use of this data set was to get the latitude and longitude of monitoring stations. These geographic coordinates are fundamental for spatial clustering, hotspot detection and mapping visualisation. These were subsequently linked to the main DEFRA data to enable space-related analysis of pollution.

second dataset

```python
from google.colab import files
files.upload() #kaggle.json
```

```
Choose files No file chosen     Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving kaggle.json to kaggle.json
{'kaggle.json': b'{"username":"manikantakesana","key":"7b56e49610c52bcbc3fccebdbc6312d4"}'}
```

```python
import os

os.makedirs('/root/.kaggle', exist_ok=True)
os.rename('kaggle.json', '/root/.kaggle/kaggle.json')
os.chmod('/root/.kaggle/kaggle.json', 600)
```

```
!kaggle datasets download -d airqualityanthony/uk-defra-aurn-air-quality-data-2015-2023
```

```
Dataset URL: https://www.kaggle.com/datasets/airqualityanthony/uk-defra-aurn-air-quality-data-2015-2023
License(s): other
uk-defra-aurn-air-quality-data-2015-2023.zip: Skipping, found more recently modified local copy (use --force to force download)
```

```
!unzip uk-defra-aurn-air-quality-data-2015-2023.zip
```

```
Archive:  uk-defra-aurn-air-quality-data-2015-2023.zip
replace AURN_2015_2023.csv? [y]es, [n]o, [A]ll, [N]one, [r]ename: A
  inflating: AURN_2015_2023.csv      A
```

Figure 2: Acquisition of the Secondary Air Quality Dataset Using Kaggle API

```python
import pandas as pd

df = pd.read_csv('AURN_2015_2023.csv')
df.head()
```

	Unnamed: 0	site	code	date	co	nox	no2	no	o3	so2	...	v10	v2.5	nv10	nv2.5	ws	wd	air_temp	latitude	longitude	site_type
0	1	Aberdeen Erroll Park	ABD9	2021-10-01 00:00:00	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	57.1574	-2.09477	Urban Background
1	2	Aberdeen Erroll Park	ABD9	2021-10-01 01:00:00	NaN	18.54277	14.25911	2.79374	35.32389	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	57.1574	-2.09477	Urban Background
2	3	Aberdeen Erroll Park	ABD9	2021-10-01 02:00:00	NaN	13.90708	12.77317	0.73952	30.23486	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	57.1574	-2.09477	Urban Background
3	4	Aberdeen Erroll Park	ABD9	2021-10-01 03:00:00	NaN	15.88038	13.32908	1.66392	21.10453	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	57.1574	-2.09477	Urban Background
4	5	Aberdeen Erroll Park	ABD9	2021-10-01 04:00:00	NaN	20.20285	18.26575	1.26334	16.41463	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	57.1574	-2.09477	Urban Background

5 rows × 22 columns

Figure 3: Preview of the UK DEFRA Air Quality Dataset After Loading into the Analysis Environment.

Observation

Geographical coordinates of air quality monitoring sites in the United Kingdom. These coordinates will be available for clustering and hotspot identification in subsequent steps of the project.

2.3 Dataset Summary

Table 1: summarises the datasets used in this research.

Dataset	Source	Purpose
Air Quality Dataset (airQuality_2024_data.csv)	DEFRA UK-AIR	Exploratory analysis, preprocessing, feature engineering, and machine learning model development
AURN Dataset (AURN_2015_2023.csv)	Kaggle	Geographic coordinate extraction, spatial clustering, hotspot detection, and map visualisation

These two datasets together not only provide geographic information but also environmental information, which allows a comprehensive analysis of the air pollution patterns and the development of prediction and hotspot detection models.

3. Dataset Overview

3.1 Dataset Structure

Following data collection, an exploratory analysis of the main air quality data was completed to gain an understanding of the structure and content of the data. The data comprises 10 attributes for 1048,575 records of pollutant measurements from monitoring stations within different regions of the United Kingdom. Individual records represent a measure of a specific pollutant at a specific time and place. The data contains the temporal data, pollutant data, monitoring station data and geographical classification. There are many records, making the data appropriate for environmental analysis and machine learning applications.

Observation

The dataset contains a substantial volume of observations, providing sufficient data for identifying pollution patterns and supporting future predictive modelling tasks.

3.2 Dataset Attributes

The primary dataset consists of ten attributes, each providing specific information about pollutant measurements and monitoring stations.

Table 2: Description of Attributes in the Primary Air Quality Dataset

Attribute	Description
StartTime	Start time of pollutant measurement.
EndTime	End time of pollutant measurement
Status	Validation status of measurement
Unit	Measurement unit of pollutant concentration
Value	Measured pollutant concentration
PollutantName	Name of the pollutant
SiteName	Monitoring station name
SiteType	Type of monitoring station
Region	Geographic region within the UK
Country	Country where the station is located

These attributes provide both environmental and contextual information required for analysing pollution behaviour across different locations and periods.

Observation

The dataset contains a combination of temporal, environmental, and location-based attributes, enabling comprehensive analysis of air quality conditions.

3.3 Pollutants Available in the Dataset

The data set includes measurements for a number of key air pollutants, which are measured at sites by the DEFRA UK-AIR network. The pollutants are used as indicators of air quality and environmental health.

The main pollutants available include:

- Nitrogen Dioxide (NO_2)
- Particulate Matter (PM10)
- Ozone (O_3)
- Sulphur Dioxide (SO_2)

For this research, Nitrogen Dioxide (NO_2) is chosen as the target pollutant, as it is one of the pollutants which poses major threats to human health and has a close correlation with traffic-related pollutants and urban air quality problems.

Observation

The dataset contains multiple pollutant categories, enabling comparative analysis of pollution levels across different pollutant types.

3.4 Initial Data Characteristics

The dataset was initially inspected and found to include pollutant data measurements from many monitoring stations in various parts of England. The measurements are carried out hourly, thus offering detailed temporal information for environmental analysis. The data is a combination of categorical and numerical. PollutantName, SiteName, Region and SiteType are categorical variables, and pollutant concentration values are numeric. This can enable the descriptive analysis and predictive modelling that can follow later in the project.

Observation

The structure of the dataset is well organised, and it has enough temporal, spatial and environmental data to analyse exploratory and to develop the machine learning.

4. Exploratory Data Analysis (EDA)

The exploratory data analysis (EDA) was performed to gain insight into the characteristics, distribution, and pattern of the air quality data before the data was preprocessed and predictive modelling was performed. The analysis was based on pollutant frequencies, pollutant value distributions, temporal pollution trends and monitoring station activity. To gain insights into the UK air quality behaviour across monitoring stations, the important patterns were identified using visualisation techniques.

4.1 Distribution of Pollutant Values

A random sample of pollutant concentration values from the data set was used to create a histogram. This analysis was conducted to gain insight into the distribution of the pollutant measurements as a whole and to see if there was any skewness or extreme values.

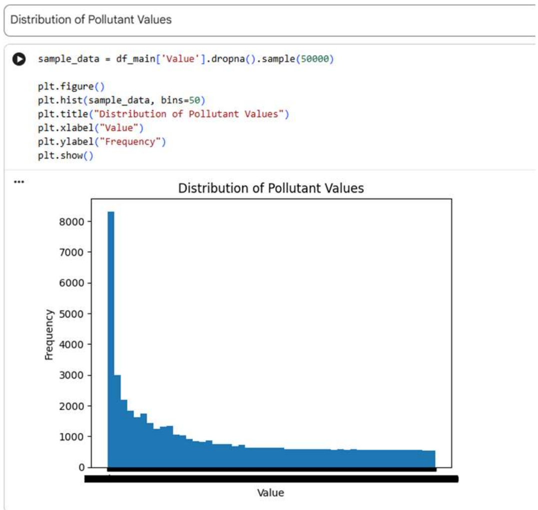

Figure 4: Distribution of Pollutant Concentration Values in the Air Quality Dataset

Interpretation

The values of pollutant concentration are not normally distributed, as indicated by the histogram. Many observations occur at low concentrations; the frequency tapers off as the pollutant increases. This points towards a positive skewness, with most measurements of air quality indicating relatively low concentrations of air pollution, and a smaller proportion pointing to high pollution events. If there are high values in the observations, the data indicate that sometimes the pollution levels are high and could be related to other activities like traffic changes, industrial emissions, or bad weather. It's important to understand this distribution because the performance of machine learning models can be affected by skewed data, and the skewed data may need to be processed later on.

4.2 Pollutant Frequency Analysis

A bar chart was drawn in order to explore the frequencies of the various measurements of pollution in the data. This analysis can be used to determine the percentage of observations made by which pollutant.

Pollutant Frequency Analysis

```
df_main['PollutantName'].value_counts().plot(kind='bar')
plt.title("Pollutant Frequency")
plt.xticks(rotation=45)
plt.show()
```

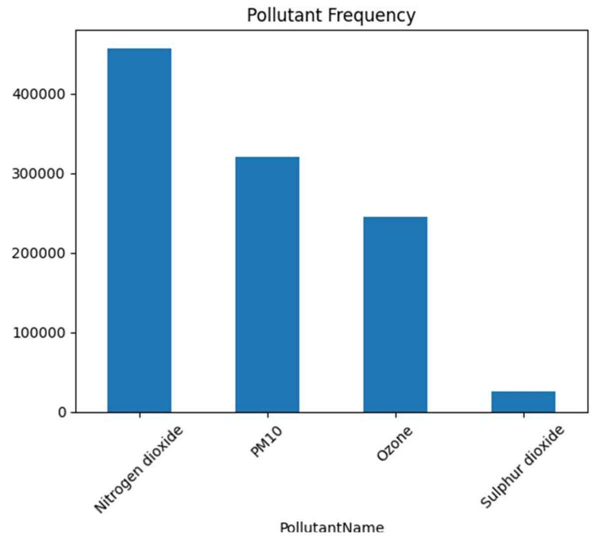

Figure 5: Frequency Distribution of Pollutants in the Air Quality Dataset

Interpretation

The results displayed indicate that the number of observations in the database for Nitrogen Dioxide (NO_2) is highest, followed by PM10 and Ozone measurements. There are fewer observations of sulphur dioxide (SO_2). The high proportion of NO_2 measurements justifies its use as the main target variable for this research. NO_2 is the largest proportion of records available, making it a good basis for statistical analysis and predictive modelling. The difference in observation numbers is also accounted for by differences in the priorities and frequency with which pollutants are monitored from station to station.

4.3 Average Pollution Over Time

The average concentration for each timestamp was estimated to examine the temporal distribution of pollution, and a line chart was used to illustrate this. This analysis gives information on variations in pollution levels over time.

Convert Value for Visualization ONLY

```
df_main.groupby('StartTime')['Value'].mean().plot()
plt.title("Average Pollution Over Time")
plt.xlabel("Time")
plt.ylabel("Average Value")
plt.show()
```

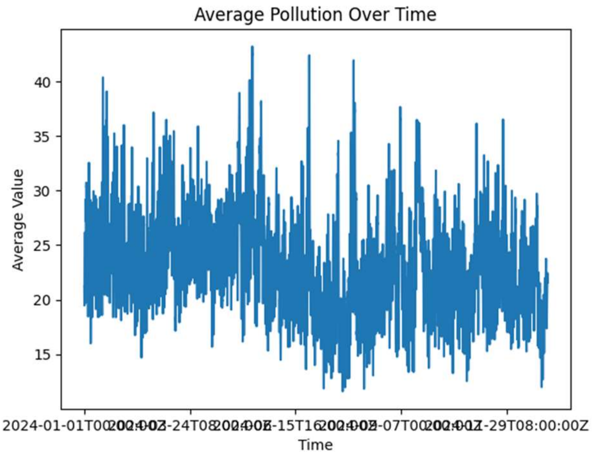

Figure 6: Temporal Trend of Average Pollutant Concentrations

Interpretation

The plot of the time series shows significant fluctuations in average pollution levels over time. Pollution levels are constantly changing, suggesting that air quality is subject to dynamic environmental and anthropogenic factors. There are a number of peaks that can be seen in the average concentrations of pollutants that are much higher than the norm. The peaks may represent times when traffic, industries or meteorological conditions have increased. The same is true for low concentration times, which indicate better air quality. These fluctuations add to the uncertainty of air pollution data as it is time-dependent and emphasise the need to consider temporal data when modelling air pollution.

4.4 Pollutant Distribution Comparison

A boxplot was created to compare the distribution of pollutant concentration between different pollutant categories. This visualisation shows information about central tendency, variability, and outliers.

Pollutant-wise Comparison

```
df_main.boxplot(column='Value', by='PollutantName')
plt.xticks(rotation=45)
plt.title("Pollutant Distribution")
plt.suptitle("")
plt.show()
```

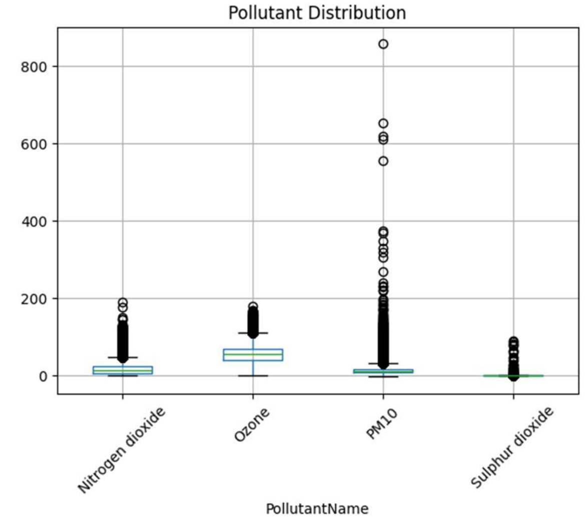

Figure 7: Comparative Distribution of Pollutant Concentrations by Pollutant Type

Interpretation

The box plot shows that the distributions of concentration are significantly different between pollutants. The median values for O_3 are the highest, whereas Sulphur Dioxide shows generally lower concentrations. There is high variability in PM10 and a large number of extreme outliers. Many outliers are present, signifying that pollution events sometimes exceed the normal levels. This is typical in environmental data sets and could be extreme pollution events. The differences in the pollutants highlight the complexity of air quality measurements and thus the requirement for further analysis of these measurements before predictive modelling.

4.5 Top Monitoring Stations Analysis

A bar chart was created to determine the number of observations from each monitoring station in the data set. The knowledge of the activity at the station is useful for assessing data coverage and locating the station's key stations.

```
df_main['SiteName'].value_counts().head(10).plot(kind='bar')
plt.title("Top Monitoring Stations")
plt.xticks(rotation=45)
plt.show()
```

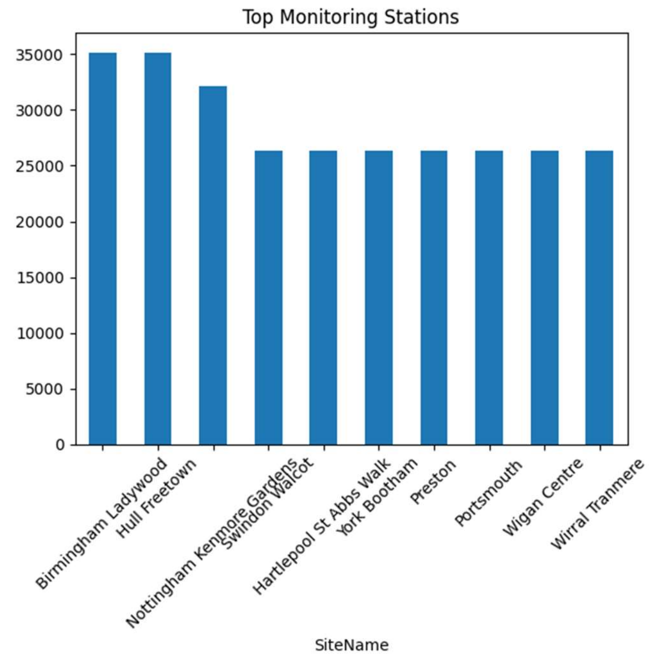

Figure 8: Top Air Quality Monitoring Stations by Number of Recorded Observations

Interpretation

The analysis shows that the number of observations was greatest at stations like Birmingham Ladywood, Hull Freetown and Nottingham Kenmore Gardens. Other stations, such as Hartlepool St Abbs Walk, York Bootham, Wigan Centre, Wirral Tranmere and Portsmouth, also provided significant data. These sites had relatively high observation counts, indicating regular monitoring and a high level of data available. These stations are important for providing the representation of air quality in various urban settings. The distribution of monitoring stations in several regions allows for the enhancement of the representativeness of the monitoring network and thus enables more reliable environmental analysis.

4.6 Summary of EDA Findings

An exploratory analysis proved useful in understanding the data structure and properties of the air quality data. The results indicated that the concentrations of pollutants were skewed and had a number of extreme observations. Nitrogen Dioxide (NO_2) was determined as the most common pollutant measured, and consequently, it was selected as the main target variable in this research. This temporal analysis showed that the pollution levels had been continuously fluctuating with time, thus emphasising the dynamism of the air quality data. The boxplot analysis showed large variations between the pollutant types, and the monitoring station analysis showed a wide geographical spread throughout the United Kingdom. The results obtained thus provide an important basis for the next steps in the project, which include data preprocessing, feature engineering, and the development of machine learning models for forecasting NO_2 concentrations.